

W10: Collinearity, Leverage and Influential Points

Tuesday, September 29, 2026

1 Collinearity

1.1 Perfect and Near Collinearity

Perfect collinearity: $X^T X$ is singular.

Near collinearity: $X^T X$ is nearly singular.

$$\Leftrightarrow |X^T X| \approx 0$$

$$\Leftrightarrow \text{at least one of the eigenvalues of } X^T X \text{ is very close to 0.}$$

Even if $X^T X$ is singular, $P(X)$ is uniquely defined.

$$\mathbf{e} = (I - P(X))\mathbf{y}, \quad \hat{\mathbf{y}}_{OLS} = P(X)\mathbf{y}, \quad \text{RSS} = \mathbf{y}^T (I - P(X))\mathbf{y}.$$

$\hat{\beta}_{OLS}$ is **NOT** uniquely defined when $X^T X$ is singular.

1.2 Illustrative Example

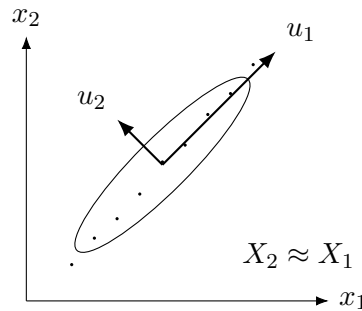


Figure 1: Near-collinear covariates x_1 and x_2 .

$$X^T X = d_1 \mathbf{u}_1 \mathbf{u}_1^T + d_2 \mathbf{u}_2 \mathbf{u}_2^T, \quad \mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

$$d_1 \gg d_2, \quad d_2 \approx 0.$$

Consider the three models:

$$Y = \alpha_0 + \beta_1 x_1 + \varepsilon_1 \Rightarrow \hat{\beta}_1 \approx \hat{\beta}_2,$$

$$Y = \alpha_1 + \beta_2 x_2 + \varepsilon_2,$$

$$Y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \approx \alpha + \underbrace{(\beta_1 + \beta_2)}_{\beta} x_1 + \varepsilon.$$

Normal equations:

$$(X^T X)\hat{\beta} = X^T \mathbf{y}.$$

Writing the normal equations in the $\{\mathbf{u}_1, \mathbf{u}_2\}$ basis:

$$\frac{d_1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (\hat{\beta}_1 + \hat{\beta}_2) + \frac{d_2}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} (\hat{\beta}_1 - \hat{\beta}_2) = X^T \mathbf{y}.$$

$$\Rightarrow \begin{bmatrix} d_1(\hat{\beta}_1 + \hat{\beta}_2) + d_2(\hat{\beta}_1 - \hat{\beta}_2) \\ d_1(\hat{\beta}_1 + \hat{\beta}_2) - \underbrace{d_2(\hat{\beta}_1 - \hat{\beta}_2)}_{\approx 0} \end{bmatrix} = 2X^T \mathbf{y}.$$

$$\Rightarrow \hat{\beta}_1 + \hat{\beta}_2 = k_0 \quad (\text{approx}).$$

$$\hat{\beta}_1 = 1000 \quad \hat{\beta}_2 = -1000 + k_0 \quad \hat{\beta}_1 - \hat{\beta}_2 = 2000 - k_0$$

OR

$$\hat{\beta}_1 = -1000 \quad \hat{\beta}_2 = 1000 + k_0 \quad \hat{\beta}_1 - \hat{\beta}_2 = -2000 - k_0$$

$\text{Var}(\hat{\beta}_1 - \hat{\beta}_2)$ is **large**.

1.3 Variance Inflation under Near Collinearity

Suppose

$$X^T X = d_1 \mathbf{u}_1 \mathbf{u}_1^T + \cdots + d_{p+1} \mathbf{u}_{p+1} \mathbf{u}_{p+1}^T.$$

Some of the d_j 's are very close to 0 [*near collinearity*].

$$\left[(\bar{X} \bar{X}) = \frac{1}{d_1} \mu_1 \mu_1^T + \cdots + \frac{1}{d_{p+1}} \mu_{p+1} \mu_{p+1}^T \right]$$

$$\text{Var}(\mathbf{a}^T \hat{\beta}_{OLS}) = \sigma^2 \mathbf{a}^T (X^T X)^{-1} \mathbf{a} = \sigma^2 \sum_{j=1}^{p+1} \frac{1}{d_j} (\mathbf{a}^T \mathbf{u}_j)^2 = \sigma^2 \sum_{j=1}^{p+1} \frac{1}{d_j} \alpha_j^2.$$

Note: $\alpha_j^2 \geq 0$ and $\sum \alpha_j^2 = \|\mathbf{a}\|^2$ ($\because \mathbf{a}^T U U^T \mathbf{a} = \mathbf{a}^T \mathbf{a}$).

If $d_j \approx 0$ for $j = j_0$, then $\frac{1}{d_j}$ is very large.

Then the j_0 -th summand, $\frac{1}{d_{j_0}} \alpha_{j_0}^2$, is also very large provided $\alpha_{j_0} = \mathbf{a}^T \mathbf{u}_{j_0}$ is not close to zero, i.e., $\mathbf{a}$ is **NOT** (nearly) orthogonal to $\mathbf{u}_{j_0}$.

1.4 Detecting Collinearity

(1) Condition Number

$$K(X) = \sqrt{\frac{\lambda_{\max}(X^T X)}{\lambda_{\min}(X^T X)}} = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)}.$$

(2) Variance Inflation Factor, $j = 1, \dots, p+1$:

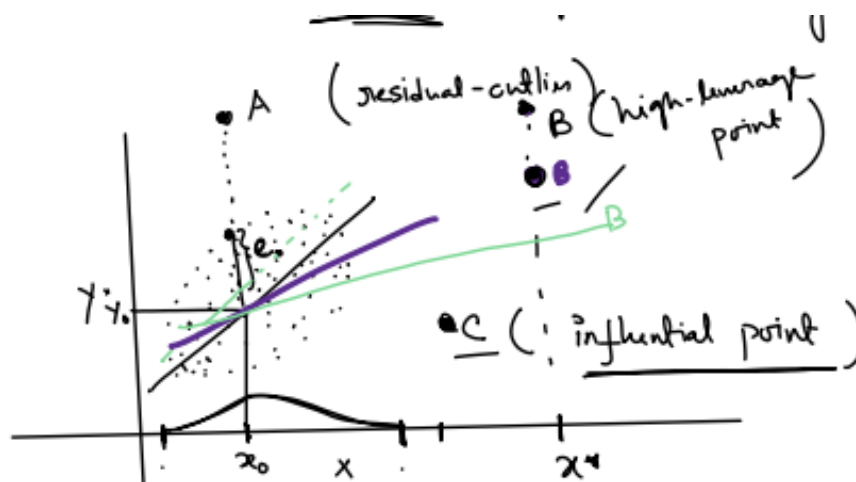
$$\text{VIF}_j = \frac{\|x_j\|^2}{x_j^T (I - P(x_{-j})) x_j} > 1,$$

since

$$\text{Var}(\hat{\beta}_{OLS,j}) = \sigma^2 x_j^T (I - P(x_{-j})) x_j.$$

If we perturb $\mathbf{y}$ a bit, then $\hat{\beta}_{OLS}$ changes drastically. (**PS**)

2 Outliers

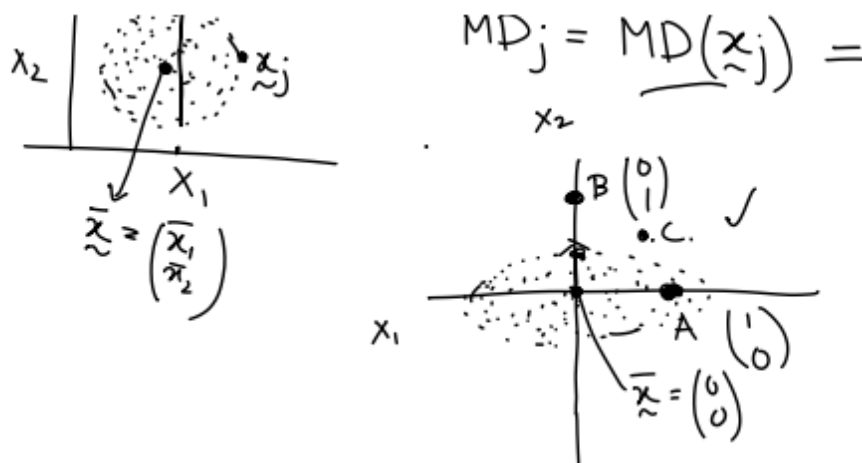


Outliers: residual-outlier, high-leverage and influential points.

2.1 Mahalanobis Distance (MD)

$$MD = \sqrt{(\mu_2 - \mu_1)^T \Sigma^{-1} (\mu_2 - \mu_1)}.$$

Suppose $\mathbf{x}_1, \dots, \mathbf{x}_n$ are the data points corresponding to the covariates.



$$MD_j = MD(\mathbf{x}_j) = \sqrt{(\mathbf{x}_j - \bar{\mathbf{x}})^T S_x^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})}.$$

$$S_x = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

2.2 The Hat Matrix

$$\hat{\mathbf{y}}_{OLS} = P(X)\mathbf{y}, \quad P(X) = ((p_{ij})).$$

i -th observation:

$$y_{OLS,i} = \mathbf{e}_i^T \hat{\mathbf{y}}_{OLS} = \mathbf{e}_i^T P(X) \mathbf{y} = p_{i1}y_1 + \cdots + p_{in}y_n = p_{ii}y_i + \sum_{j \neq i} p_{ij}y_j.$$

Derivation of p_{ii} :

$$\begin{aligned} p_{ii} &= \mathbf{e}_i^T P(X) \mathbf{e}_i \\ &= \mathbf{e}_i^T \left[\frac{1}{n} \mathbf{1} \mathbf{1}^T + P(Z_c) \right] \mathbf{e}_i, \quad \text{where } Z_c = [\mathbf{x}_1 - \bar{x}_1 \mathbf{1}, \dots, \mathbf{x}_p - \bar{x}_p \mathbf{1}] \\ &= \frac{1}{n} + \mathbf{e}_i^T Z_c S_x^{-1} Z_c^T \mathbf{e}_i / n, \quad P(Z_c) = Z_c (Z_c^T Z_c)^{-1} Z_c^T \\ &= \frac{1}{n} \left\{ 1 + [(x_{1i} - \bar{x}_1) \cdots (x_{pi} - \bar{x}_p)] S_x^{-1} \begin{bmatrix} (x_{1i} - \bar{x}_1) \\ \vdots \\ (x_{pi} - \bar{x}_p) \end{bmatrix} \right\} \end{aligned}$$

where

$$\underbrace{Z_c^T Z_c}_{p \times p} = \left(\left(\sum_{k=1}^n (x_{ik} - \bar{x}_i)(x_{jk} - \bar{x}_j) \right) \right)_{\substack{i=1, \dots, p \\ j=1, \dots, p}} = n S_x,$$

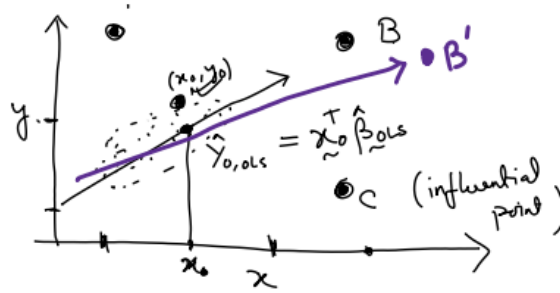
$$P(Z_c) = Z_c (n S_x)^{-1} Z_c^T = n^{-1} Z_c S_x^{-1} Z_c^T.$$

$$p_{ii} = \frac{1}{n} \{ 1 + (\mathbf{x}_i - \bar{\mathbf{x}})^T S_x^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}) \}, \quad (i\text{-th observation on the covariates})$$

$$p_{ii} = \frac{1}{n} \{ 1 + \text{MD}_i^2 \}.$$

3 Leverage and Influence

$$\hat{\mathbf{y}}_{obs} = X \hat{\beta}_{OLS} = X (X^T X)^{-1} X^T \mathbf{y} = P(X) \mathbf{y}.$$



i -th observation:

$$\mathbf{e}_i^T \hat{\mathbf{y}}_{obs} = \hat{y}_{OLS,i} = \mathbf{e}_i^T P(X) \mathbf{y} = \mathbf{p}_i \cdot \mathbf{y}, \quad \mathbf{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i\text{-th},$$

$$= p_{i1}y_1 + \cdots + p_{in}y_n, \quad \text{where } \mathbf{p}_i : i\text{-th row of } P(X).$$

$$\hat{y}_{obs,i} = p_{ii}y_i + \sum_{j \neq i} p_{ij}y_j.$$

3.1 Key Facts about $P(X)$

Fact (i):

$$p_{ii} = \frac{1}{n} [1 + \text{MD}_i^2], \quad p_{ii} \geq \frac{1}{n},$$

$$\text{MD}_i^2 = (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^T S_x^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where $\mathbf{x}_{(i)}$: i -th observation, $\bar{\mathbf{x}} = [\bar{x}_1, \dots, \bar{x}_p]^T$.

Fact (ii):

$$P(X)^2 = P(X).$$

$$p_{ii} = \mathbf{p}_i^T \mathbf{p}_i = p_{i1}^2 + \dots + p_{in}^2 \quad \text{--- } (*)$$

$$p_{ii}^2 \leq p_{ii} \Rightarrow p_{ii} \leq 1.$$

$$\frac{1}{n} \leq p_{ii} \leq 1.$$

$$p_{ii}(p_{ii} - 1) + \sum_{j \neq i} p_{ij}^2 = 0.$$

From (*) we get: if p_{ii} is large, i.e., close to 1, then $\{p_{i1}, \dots, p_{i,i-1}, p_{i,i+1}, \dots, p_{in}\}$ are small.

Thus, a **high-leverage point** has **higher influence** on the regression line.

3.2 High-Residual Point but Not High-Leverage Point

In this case, the residuals can identify such a point quite well.

To see this, suppose y_n is a high-residual point:

$$\left. \begin{aligned} y_i &= \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n-1 \\ y_n &= \mathbf{x}_n^T \boldsymbol{\beta} + \delta + \varepsilon_n, \quad |\delta| \text{ is large.} \end{aligned} \right\} \Rightarrow \mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\Delta} + \boldsymbol{\varepsilon},$$

$$\boldsymbol{\Delta} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \delta \end{bmatrix}, \quad \boldsymbol{\varepsilon} \sim N(0, \sigma^2 I).$$

Residual:

$$\mathbf{e} = (I - P(X))\mathbf{y} = (I - P(X))\boldsymbol{\Delta} + (I - P(X))\boldsymbol{\varepsilon}.$$

$$E(\mathbf{e}) = (I - P(X))\boldsymbol{\Delta} = Q(X)\boldsymbol{\Delta}.$$

n -th expected residual is:

$$E(\mathbf{e}) = \begin{bmatrix} E(e_1) \\ \vdots \\ E(e_n) \end{bmatrix} = \delta \cdot \mathbf{q}_{\cdot n}, \quad \mathbf{q}_{\cdot n} : \text{last column of } Q(X).$$

With

$$Q(X) = I - P(X) = \begin{bmatrix} 1 - p_{11} & -p_{12} & \cdots & -p_{1n} \\ \vdots & & & \vdots \\ -p_{n1} & -p_{n2} & \cdots & 1 - p_{nn} \end{bmatrix},$$

$$E(\mathbf{e}) = \begin{bmatrix} -\delta p_{1n} \\ \vdots \\ -\delta(1 - p_{nn}) \end{bmatrix} \approx \begin{bmatrix} \delta \left(1 - \frac{p}{n}\right) \\ \vdots \\ \delta \left(1 - \frac{p}{n}\right) \end{bmatrix} \quad (\text{e.g. } n = 10, p = 4, \delta = 5).$$

If $(\mathbf{x}_n, y_n)$ is **not** a high-leverage point, then p_{nn} is small.

$$\sum p_{ii} = p + 1. \quad \text{Usually } p_{ii} \sim \frac{p+1}{n}.$$

Similarly, it can be shown that

$$p_{ij} \sim \frac{1}{n} \left[1 - \frac{p}{n-1} \right].$$

Therefore, from the residual plot, a high-residual outlier **can be identified**, as long as it is **not also** a high-leverage point.

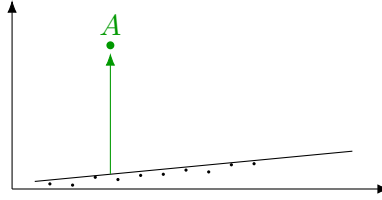


Figure 2: High-residual outlier A , identified via the residual plot.

3.3 Remark on Robustness

Remark: While identifying a high-leverage point by MD_i or p_{ii} , note that both $\bar{\mathbf{x}}, S_x$ are affected by an extreme observation. So, we have to replace $\bar{\mathbf{x}}, S_x$ by some **robust estimators** of location and scale.

Thus, we may consider a robust version of MD: $(\boldsymbol{\mu}_x, \Sigma_x)$.

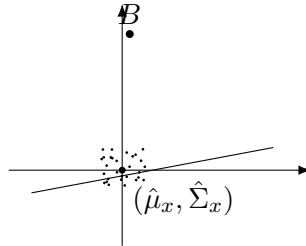


Figure 3: Robust location/scale estimate affected by an extreme observation B .

4 Robust Mahalanobis Distance

$$\widetilde{MD}_i^2 = (\mathbf{x}_i - m(X))^T C(X)^{-1} (\mathbf{x}_i - m(X)),$$

where $m(X)$ is a **robust location estimate** (replacing $\bar{\mathbf{x}}$) and $C(X)$ is a **robust scale estimate** (replacing S_x).

4.1 Choices

- $m(X)$ and $C(X)$ can be the **trimmed mean and variance**, i.e., $\bar{\mathbf{x}}_{(n-h)}$ and $S_{x,n-h}$ (i.e., $\bar{\mathbf{x}}$ and S_x calculated using a subset of $\{\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(n)}\}$ of $(n-h)$ observations).
- Which $(n-h)$ observations to retain?

Minimum Covariance Determinant (MCD) method: Consider that subset for which $S_{x,n-h}$ has the **lowest determinant**.

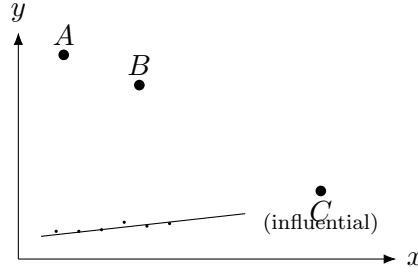


Figure 4: Candidate outlier/influential points A, B, C.

$$\mathcal{E}_k = \{\mathbf{x} : (\mathbf{x} - \mathbf{m})^T C^{-1} (\mathbf{x} - \mathbf{m}) \leq k\}, \quad \text{vol}(\mathcal{E}_k) \propto |\det(C)|.$$

A point is **influential** if the regression line including that point is very different from the one excluding that point.

5 Influential Points

To identify such a point, one must consider some function of $(\hat{\beta}_{OLS} - \hat{\beta}_{OLS,(-i)})$.

Result:

$$\hat{\beta}_{OLS} - \hat{\beta}_{OLS,(-i)} = (X^T X)^{-1} \mathbf{x}_i \left(\frac{e_i}{1 - p_{ii}} \right).$$

5.1 Proof

$\hat{\beta}_{OLS,(-i)}$ = OLS estimate of β considering $(n-1)$ observations $\{1, \dots, i-1, i+1, \dots, n\}$.

$$\underbrace{X_{(-i)}}_{(n-1) \times (p+1)} = \begin{bmatrix} \mathbf{x}_{(1)}^T \\ \vdots \\ \mathbf{x}_{(i-1)}^T \\ \mathbf{x}_{(i+1)}^T \\ \vdots \\ \mathbf{x}_{(n)}^T \end{bmatrix}.$$

$$X^T X = \begin{bmatrix} \mathbf{x}_{(1)} & \cdots & \mathbf{x}_{(i)} & \cdots & \mathbf{x}_{(n)} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{(1)}^T \\ \vdots \\ \mathbf{x}_{(i)}^T \\ \vdots \\ \mathbf{x}_{(n)}^T \end{bmatrix} = \mathbf{x}_{(1)} \mathbf{x}_{(1)}^T + \cdots + \mathbf{x}_{(n)} \mathbf{x}_{(n)}^T = X_{(-i)}^T X_{(-i)} + \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T.$$

Woodbury Matrix Identity:

$$(A + \mathbf{u} \mathbf{v}^T)^{-1} = A^{-1} - A^{-1} \mathbf{u} (1 + \mathbf{v}^T A^{-1} \mathbf{u})^{-1} \mathbf{v}^T A^{-1}.$$

Hence,

$$X_{(-i)}^T X_{(-i)} = X^T X - \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T,$$

$$\begin{aligned} \left(X_{(-i)}^T X_{(-i)} \right)^{-1} &= \left(X^T X - \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T \right)^{-1} \\ &= (X^T X)^{-1} - (X^T X)^{-1} \mathbf{x}_{(i)} \left(-1 + \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)} \right)^{-1} \mathbf{x}_{(i)}^T (X^T X)^{-1} \\ &= (X^T X)^{-1} + \frac{(X^T X)^{-1} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T (X^T X)^{-1}}{1 - \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)}}. \end{aligned}$$

Also,

$$\begin{aligned} X^T \mathbf{y} &= \begin{bmatrix} \mathbf{x}_{(1)} & \cdots & \mathbf{x}_{(n)} \end{bmatrix} \mathbf{y} = y_1 \mathbf{x}_{(1)} + \cdots + y_n \mathbf{x}_{(n)}, \\ X_{(-i)}^T \mathbf{y} &= X^T \mathbf{y} - y_i \mathbf{x}_{(i)}. \\ \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)} &= p_{ii}. \end{aligned}$$

Therefore,

$$\begin{aligned} \hat{\beta}_{OLS,(-i)} &= \left(X_{(-i)}^T X_{(-i)} \right)^{-1} X_{(-i)}^T \mathbf{y} \\ &= \left[(X^T X)^{-1} + \frac{(X^T X)^{-1} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T (X^T X)^{-1}}{1 - \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)}} \right] (X^T \mathbf{y} - y_i \mathbf{x}_{(i)}). \end{aligned}$$

where

$$\underbrace{X}_{n \times (p+1)}, \quad \underbrace{X_{(-i)}}_{(n-1) \times (p+1)}, \quad X(X^T X)^{-1} X^T = P(X).$$